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HateBertBN: a hybrid transformer based model for Bangla hate speech detection across various social contexts

Tanvir Azhar^{1*}, Tahsin Mahmud¹, Muhammad Asif Hasan¹, Mohammed Nazim Uddin¹ and Seung-Bo Park²

*Correspondence:

Tanvir Azhar

azhar.t@eastdelta.edu.bd

¹School of Science, Engineering & Technology, East Delta University, Chattogram 4209, Bangladesh

²Department of Software Convergence Engineering, Inha University, Michuhol-gu, Incheon 22212, Republic of Korea

Abstract

The widespread use of online social media platforms has amplified the importance of efficient hate speech detection, especially in low-resource languages like Bengali. While traditional machine learning approaches show promise, deep learning is more effective in capturing the nuanced context of hate speech. Current challenges include a lack of diverse datasets and models capable of context-sensitive detection. To address these, we introduce HateCorpBN-XL, the largest labeled Bengali hate speech dataset to date, containing 65,251 comments across five categories: political (PoHS), religious (ReHS), misogynistic (MisoHS), slander (SlaHS), and xenophobic (XenHS). We also propose HateBertBN, a hybrid transformer-based model combining BanglaBERT embeddings with three neural network fusion strategies using CNN, LSTM, and MLP. We evaluate our approach on two tasks, Task-1: detecting hate speech in Bengali text classifying it as hateful or non-hateful and Task-2: categorizing hateful content into five distinct classes. For Task-1, all HateBertBN variants outperformed current transformer models, achieving an accuracy of 0.92 and a weighted F1-score of 0.92. In Task-2, the HateBertBN-MLP and HateBertBN-CNN variants achieved a notable 0.90 accuracy and weighted F1-score of 0.90, surpassing M-BERT, Distil-M-BERT, BanglaBERT, and XLM-R-Base. Although HateBertBN-LSTM performed slightly lower overall, it achieved strong F1-scores in the ReHS (0.93) and XenHS (1.00) categories. Overall, our hybrid model outperforms state-of-the-art approaches in both tasks, demonstrating its effectiveness and robustness.

Keywords Hate speech, Social media, Large language models, BERT, XLM-R, CNN, LSTM, MLP

1 Introduction

With the growth of the internet and mobile technology over the past decade, social media platforms like Facebook, Instagram, and Twitter have seen significant expansion. These platforms enable users to share their thoughts and ideas visibly with the world. Unfortunately, many people also use the internet to spread violent, hateful, or threatening messages. Deirmenjian et al. [1] describe hate speech (HS) as any public communication that promotes hatred or supports violence against individuals or groups based



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on race, religion, gender, nationality, color, sexual orientation, or other characteristics. The main goal of the online world is to help people feel safe and connected. However, maintaining community standards is becoming more difficult due to the rapid growth of users. Statistics show that teenagers spend more time on social networking sites, and nearly 39.9% of students have reported being cyberbullied [2]. Therefore, it is crucial for government entities, law enforcement, social security agencies, and social media companies to effectively remove accounts that spread hate speech from these platforms. Many studies have aimed to automate the detection of hate speech on social media using natural language processing (NLP), machine learning, data mining, content extraction and retrieval, and text mining [3, 4]. The most popular approach for text classification problems is machine learning, which uses labeled samples as training data to learn the correlations between texts and predict specific outputs for given inputs [5, 6]. However, deep learning, a subset of machine learning that relies on deep artificial neural networks, has recently shown promising results in complex text classification problems [7, 8].

Inspired by this, our study proposes a hybrid deep learning-based transformer model for classifying Bengali hate speech to facilitate more precise and efficient automation. It's important to note that the development of social media monitoring and analysis tools is common for high-resource languages like English but is rare for low-resource languages such as Bengali. These languages lack essential linguistic resources, such as language models, datasets, and effective preprocessing steps for various NLP tasks [9, 10].

In this work, we address the issue of Bengali hate speech on popular online social media platforms in response to national and global demands. Initially, the study focused on collecting data from various sources, which includes religious, political, gender, social or personal, geopolitical, and ethnic minority related comments to capture diverse words and linguistic characteristics. Additionally, several text data preprocessing steps were applied for data cleaning, such as tokenization, stop word removal, hashtag removal, punctuation removal, and stemming. Finally, we explored three distinct hybrid approaches—HateBertBN-CNN with convolutional layers, HateBertBN-LSTM with sequential modeling, and HateBertBN-MLP with dense neural networks, to enhance the performance of Bengali hate speech classification.

In summary, the contributions of this study are the following:

- Develop a Bengali hate speech dataset containing 30,250 hate comments with their corresponding labels and a total 65,251 comments. A hierarchical annotation schema is used to further annotate hate speech into misogynistic, xenophobic, political, religious, and slander classes. We define two core tasks:
 - *Task 1*: Detect hate speech from a given Bengali text corpus, distinguishing hate speech (HS) content from non-hateful speech (NHS).
 - *Task 2*: Perform fine-grained classification of the detected hate speech into five distinct categories: Misogynistic (MisoHS), Xenophobic (XenHS), Political (PoHS), Religious (ReHS), and Slander (SlaHS).
- Propose HateBertBN a hybrid transformer model with its three variants where we have shown the effectiveness of fusing CNN, LSTM and MLP for context-based hate speech classification. We have evaluated and compared our proposed model

with the current state-of-the-art deep learning models by validating our results using both qualitative and quantitative analysis.

2 Related works

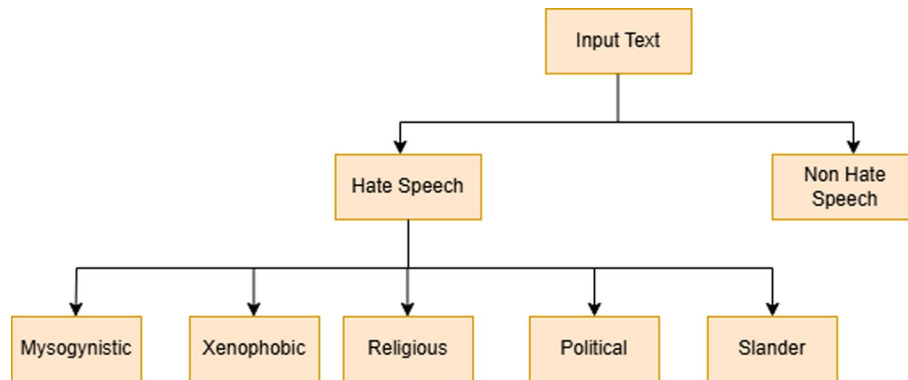
In recent years, there has been a significant amount of work in hate speech (HS) detection. Most of the studies have focused on the English language [11]. HS detection has also been explored for detecting trolling [12, 13], cyberbullying [14, 15], cyber abuse [16, 17], and toxicity [18, 19] in different languages. Some notable works have also been conducted in the Bangla language. These include both binary and multi-class classification tasks based on dataset configurations. Ghosh et al. [20] proposed a hybrid deep learning approach combining Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Convolutional Neural Network (CNN) models to effectively detect HS in Bangla social media. Their approach addressed the limitations of traditional machine learning models in handling the complex linguistic features of Bangla, a language with limited preprocessing resources such as stemmers and lemmatizers. They utilized a dataset of 30,000 social media comments collected from Facebook and YouTube, annotated through crowdsourcing. Their hybrid model, leveraging FastText embeddings, outperformed Bengali Bidirectional Encoder Representations from Transformers (BanglaBERT) and Bangla Efficiently Learning an Encoder that Classifies Token Replacements Accurately (ELECTRA), achieving an accuracy of 0.90. Das et al. [21] collected 7425 Bengali Facebook comments categorized into seven types of HS. They employed an LSTM-based encoder and an attention-based decoder mechanism. Experiments with LSTM and GRU models showed that the attention-based decoder achieved the best accuracy of 0.77 for HS category prediction. Romim et al. [22] developed a dataset of 30,000 comments collected from Facebook and YouTube, categorized into seven HS-related classes. They experimented with baseline models such as SVM, LSTM, and Bidirectional LSTM (BiLSTM) using various word embeddings, including Word2Vec, FastText, and BengFastText. Among these, the SVM model achieved the highest accuracy of 0.87. Junaid et al. [23] focused on classifying YouTube videos using machine learning and deep learning techniques. Their dataset contained 300 videos, and among various models, logistic regression achieved an accuracy of 0.9620, while the GRU-based deep learning model achieved 0.9889. Jahan et al. [24] introduced BanglaHateBERT, a pretrained BERT-based model for Bengali hateful language recognition. They collected data from Facebook and YouTube, creating a dataset of 15,000 posts across eight categories (xenophobia, racism, sexual, religious, homophobia, personal, geopolitical, and political). Their model achieved 0.943 accuracy in HS category detection and 0.931 in HS identification. Sharif et al. [25] developed a dataset containing 14,158 texts categorized into four classes: religious, political, verbal, and gendered aggression. They proposed a weighted ensemble approach incorporating Multilingual Bidirectional Encoder Representations from Transformers (mBERT), Distilled Bidirectional Encoder Representations from Transformers (DistilBERT), BanglaBERT, and Cross-lingual Language Model—RoBERTa (XLM-R) as base classifiers. Their method achieved a weighted F1-score of 0.9343 for HS identification and 0.9311 for HS categorization. Emon et al. [26] evaluated three transformer models—BanglaBERT, Bengali DistilBERT, and XLM-RoBERTa on a dataset of 44,001 Facebook comments obtained from Mendeley. Among them, XLM-RoBERTa achieved the highest accuracy of 0.85. Remon et al. [27] built a

dataset of 10,133 Bengali Facebook comments and evaluated several machine learning and deep learning models. The SVM model achieved the best performance with an accuracy of 0.8722. Romim et al. [28] introduced a dataset of 50,200 offensive comments collected from various social media platforms. They trained SVM and BiLSTM models with features such as unigram, character n-gram, BanglaFastText, MultilingualFastText, and InFormal Text (IFT)-a FastText model trained from scratch on informal text. The SVM model achieved F1-scores of 0.909 for HS identification and 0.891 for HS categorization, while BiLSTM obtained 0.775 accuracy for HS target identification. Das et al. [29] created a dataset of 10,000 Bengali tweets (5000 standard and 5,000 Romanized) and trained transformer models including mBERT, XLM-RoBERTa, Multilingual Representations for Indian Languages (MuRIL), and IndicBERT. XLM-RoBERTa achieved 0.83 accuracy on standard Bangla tweets and 0.858 on Romanized tweets, while MuRIL performed best on the combined dataset with 0.856 accuracy. Keya et al. [30] proposed G-BERT, a hybrid model combining BERT and GRU, evaluated on the Bengali Hate Speech from Social Platforms (BHSSP) dataset containing 2932 offensive and 4011 non-offensive comments in both Bengali and Banglish. Their G-BERT model achieved an accuracy of 0.91, outperforming Gradient Boosting Trees (GBT), BanglaBERT, XLM-RoBERTa, mBERT-uncased, and Mass-Conserving-LSTM (MC-LSTM). Nishat Raihan et al. [31] addressed offensive language detection in transliterated and code-mixed Bangla, introducing the TB-OLID dataset of 5000 manually annotated Facebook comments. Evaluations using BERT, RoBERTa, mBERT, XLM-RoBERTa, HateBERT, and fBERT revealed that fBERT and HateBERT performed best, achieving an F1-score of 0.72. Hossain et al. [32] developed a dataset of 6100 audio clips collected from 20 districts in Bangladesh, which is publicly available. Raihan et al. [33] explored the Violence Inciting Text Detection (VITD) task using a dataset provided by the shared task organizers, annotated into Non-Violence, Passive Violence, and Direct Violence. They employed a two-step classification approach with data augmentation, fine-tuning transformer models such as BanglaBERT, MuRIL, XLM-R, and BanglaHateBERT. The best performance was achieved by XLM-R and MuRIL with a macro F1-score of 0.74. Beyond Bangla, studies in other low-resource languages have also gained attention. Daneshfar [34] conducted sentiment analysis in Central Kurdish, introducing gold-standard and silver-standard datasets with 1185 and 4500 samples, respectively. Among several models-logistic regression, decision trees, random forest, SVM, and BiLSTM-the SVM achieved the highest accuracy of 0.61 when combining both datasets. Haq et al. [35] developed the Pashto Offensive Language Dataset (POLD) containing 34,400 labeled tweets (12,400 offensive, 22,000 non-offensive). They evaluated CNN, LSTM, GRU, BiLSTM, and Bidirectional GRU models with static word embeddings (Word2Vec, FastText, and GloVe), and also fine-tuned multilingual and monolingual transformer models (XLM-R and Ps-BERT). The Ps-BERT model achieved an F1-score of 0.94 and an accuracy of 0.9477. For multilingual populations, HS is often expressed in code-mixed languages such as Hindi-English (Hinglish). Sreelakshmi et al. [36] proposed a machine learning model for HS detection in Hinglish social media text. Using embeddings such as fastText, Word2Vec, and Doc2Vec, and SVM with an RBF kernel, their model achieved an accuracy of 0.8581, outperforming Random Forest and other baselines.

Table 1 provides a summary of previous related works on Bangla HS detection.

Table 1 Summary of previous related works on Bangla hate speech detection

Authors	Corpus size	Models	Best model	Performance
Das et al. [21]	7425	GRU, LSTM, LSTM (encoder)–attention-based decoder	LSTM (encoder)–attention-based decoder	0.7700 (Multi-class)
Romim et al. [22]	30,000	SVM, LSTM, BiLSTM with different word embeddings	SVM	0.8700 (Binary)
Sharif et al. [25]	14,158	mBERT, DistilBERT, BanglaBERT, XLM-R (weighted ensemble)	Ensemble model	0.9343 (Binary), 0.9311 (Multi-class)
Emon et al. [26]	44,001	BanglaBERT, Bengali DistilBERT, XLM-RoBERTa	XLM-RoBERTa	0.8500 (Binary)
Romim et al. [28]	50,200	SVM, BiLSTM with different embeddings	SVM (char n-gram)	0.9090 (Binary), 0.8910 (Multi-class)
Ghosh et al. [20]	30,000	Bangla ELECTRA, BanglaBERT, LSTM–GRU–CNN	LSTM–GRU–CNN	0.9000 (Binary)
Das et al. [29]	10,000	mBERT, XLM-RoBERTa, MuRIL, IndicBERT	XLM-RoBERTa	0.8560 (Joint Text, Binary)
Keya et al. [30]	6943	GBT, BanglaBERT, XLM-RoBERTa, mBERT, G-BERT	G-BERT	0.9100 (Binary)
Raihan et al. [33]	6046	BanglaBERT, MuRIL, XLM-R, BanglaHateBERT	XLM-R, MuRIL	0.7400 (Binary)
Sreelakshmi et al. [36]	10,000	SVM-Linear, SVM-RBF, Random Forest	SVM-RBF	0.8581 (Binary)

**Fig. 1** Classification schema of the proposed tasks

3 Task definition

The goal of this effort is to create a robust hate speech text classification and categorization system. The system operates on two layers. The proposed system's classification schema is shown in Fig. 1. Initially, the system identifies whether a given comment $C_i \in C$, where $C = \{C_1, C_2, \dots, C_N\}$, contains hate speech or not. This binary classification task distinguishes between hate and non-hate comments. If the comment is identified as hate speech, the system goes to the next level. At this level, the system further classifies comments identified as hate speech into one of n predefined categories, $HC = \{hc_1, hc_2, \dots, hc_n\}$, where hc_j represents a specific hate speech class.

Our investigation shows that different people classify hateful contents in distinct ways due to dialect complexities of each languages, thus it's hard to establish boundaries between these hostile text categories. Our selected categories for this study—misogyny, xenophobia, political, religious and slanderous speech can be found in other works of literature such as aggression classification [37–39], HS identification [40–42],

abuse detection [43], cyber-bullying categorization [44], misogyny, xenophobia on social media [45, 46] which helped us to define and distinguish the following groups of categories as our target classes:

- *Misogynistic (MisoHS)*: Content that promotes hatred, violence, or discrimination against individuals based on their gender, specifically targeting women by reinforcing harmful stereotypes, demeaning their worth, or advocating for their oppression.
- *Xenophobic (XenHS)*: Content that expresses hatred, discrimination, or exclusion against individuals or groups based on their nationality, ethnicity, or status as immigrants, often portraying them as dangerous, inferior, or unworthy of inclusion.
- *Political (PoHS)*: Content that incites violence, harassment, or discrimination against individuals or groups based on their political beliefs, affiliations, or party membership.
- *Religious (ReHS)*: Content that attacks or incites hatred against individuals or groups based on their religion, including derogatory statements, dehumanizing comparisons, or calls for violence against specific religious communities.
- *Slander (SlaHS)*: False or misleading statements made with the intent to damage an individual's or organization's reputation, particularly when these statements include hate-driven elements aimed at inciting discrimination or harm (Table 2).

4 Dataset development

A significant drawback in the research methods within this field is the lack of a comparative analysis of publicly available Bengali datasets. The majority of existing studies rely on privately acquired datasets, and many of these studies involve only a limited Bengali text collection. While extensive corpora for other languages are readily accessible online, it is uncommon for low-resource languages like Bengali to have access to such resources. Hence, we recognized the necessity to develop our own dataset. We developed a dataset which is larger than any other available datasets in Bengali. Our dataset development steps are presented in Fig. 2. We followed the guidelines provided by Vidgen and Derczynski [47]. A thorough examination of the dataset development process is described in the subsequent subsections.

4.1 Data acquisition

We collected a total of 65,251 aggressive and non-aggressive texts from various social networking platforms. To ensure diversity, the data were gathered from a wide range of online discussions and user interactions. The collected texts were categorized into five distinct types—Misogynistic, Xenophobic, Political, Religious, and Slander ensuring comprehensive coverage in terms of both linguistic expression and topical representation. Our source of the comments were public Facebook posts and YouTube comments

Table 2 Comparison with existing datasets

Paper	Texts	Classes
Das et al. [21]	7425	7
Sharif et al. [25]	14,158	4
Romim et al. [22]	30,000	7
Emon et al. [26]	44,001	5
Romim et al. (BD-SHS) [28]	50,200	4
HateCorpBN-XL	65,251	5

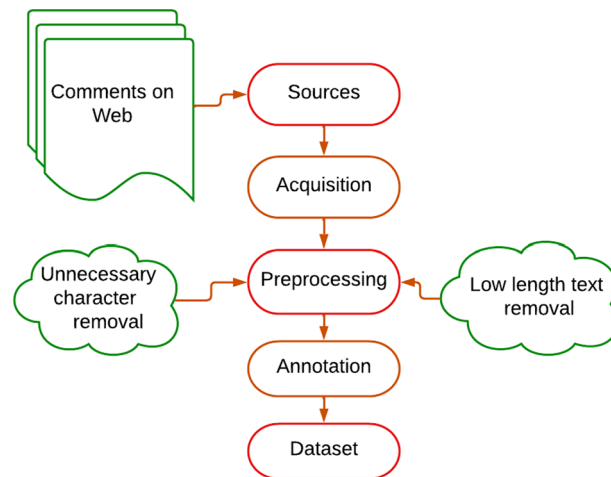


Fig. 2 Dataset development steps

Table 3 Data acquisition sources

Name	Type	Category	No. of followers/subscribers
Bangladesh Cricket : The Tigers	Facebook	Sports	15M
Prothom Alo	Facebook	Newsgroup	21M
Elias Hossain	YouTube	Public figure	4.67M
Jamuna TV	YouTube	Media	29.1M
Bangladesh Nationalist Party-BNP	Facebook	Political	3.8M
Bangladesh Awami League	Facebook	Political	4.1M
Dr. Asif Nazrul	Facebook	Political	2.2M
Basher Kella	Facebook	Political	2M
Salman Mohammad Muqtadir	Facebook	Public figure	4.2M
Shakib Khan	Facebook	Celebrity	7.4M
Pinaki Bhattacharya	Youtube	Author	3.91M
Ahmadullah	Facebook	Islamic figure	6.5M

section since these are the most popular social media platform in Bangladesh. According to social media statistics, Facebook is used by 84.72% Bangladeshi users and YouTube is used by 10.98% users [48]. The entire data collection process was conducted manually for educational and research purposes, adhering strictly to social media platform policies. The collection spanned eight months. We have observed controversial and potential hateful posts from February 15, 2024, to August 25, 2024. We have collected comments from the thread which accumulated more than 250 reactions in the social platforms. Potential aggressive text were primarily collected from more than 100 Facebook pages and YouTube channels associated with public figures, religious activists, political commentators, sports organizations, and celebrities, as these sources frequently host emotionally charged and argumentative discussions. Most frequent and popular sources are mentioned in Table 3 where only publicly accessible comment sections were considered, and no private content was accessed. In contrast, non-aggressive texts were collected from posts concerning education, sports, technology, science, and general news. The source pages or channels might contain personal information; thus, we avoided disclosing the source link. The collected comments were systematically documented, ensuring traceability and organization. No automated web-scraping tools or APIs were employed to maintain compliance with platform copyright and fair use regulations. The retrieved

comments were then further categorized by trained annotators following detailed annotation guidelines to ensure consistency and reduce subjectivity. A comment was labeled as aggressive or violence-provoking if it exhibited one or more of the following characteristics: explicit threats or calls for harm; incitement to violence or aggression; use of slurs, obscene, or derogatory language toward individuals or groups; expression of hate or hostility based on gender, religion, ethnicity, or political stance; or an overall antagonistic or inflammatory tone. Conversely, non-aggressive comments were those that were neutral, informative, or constructive in nature. Ambiguous cases were reviewed collectively by the annotation team to achieve a consensus decision. To ensure linguistic purity and consistency, we applied a two-step filtering process to retain only authentic Bengali text. During collection, annotators were instructed to focus on Bangla-written comments; however, code-mixed or Romanized (Banglish) instances were occasionally found. After the initial collection, all comments were processed using a Python script with the “langdetect” library, which automatically filtered out comments containing non-Bengali unicode characters. The remaining comments were then manually reviewed by trained annotators to exclude Romanized Bengali, mixed-language, or semantically irrelevant content.

4.2 Data pre-processing

A few pre-processing tasks were performed to give the comments some organization and completeness:

1. Stop Words Removal: Stop words are the words that people use most often in a language that do not give much meaning to a sentence. These can be discarded without affecting the meaning of the sentence. For example: “আমি”, “তুমি”, “সে” which means “I”, “you”, “he/she”.
2. Special Character Removal: Special characters such as hyphen (-), semicolon (;), comma (,), and question mark (?) were removed. We applied the Regular Expression (re) Python library for this task.
3. Basic Form Processing: The comments were processed to remove URLs, Hashtags, and mentions (@username).
4. Stemming: Stemming is a technique to reduce a word to its base word. For example, the words “আমার”, “আমি”, “আমি” all derive from the same common word “আমি”. In this way, the model will learn that all three words are used in the same context and have a similar meaning.

While transformers are robust to raw text, these preprocessing steps enhanced data consistency, reduced noise, and slightly improved training efficiency. In practice, we observed that removing special characters and normalizing text contributed positively to the model’s performance, while stop word removal and stemming had a minor but consistent effect. Table 4 shows the outcomes of the preprocessing.

4.3 Data annotation

As established in prior work [18], an essential first step in developing a HS dataset is to distinguish between HS and NHS instances. To achieve this, specific rules were established, and annotators were selected to identify HS instances. The following provides a brief overview of the annotators and the guidelines employed for annotation.

Table 4 Examples of hate comments before and after preprocessing

Before preprocessing	After preprocessing
@username শালায় এই কারণেই খেলায় ডাববা মারে!!!!	শালায় এই কারণেই খেলায় ডাববা মারে
পাপন কুত্তার বাচ্চা!!!! এই ভিডিও টা দেখছে কি???	পাপন কুত্তার বাচ্চা এই ভিডিও টা দেখছে কি
দুইচ্ছারা। প্রেম। করতে। পারে। বিয়ে। করতে। পারেনা	দুইচ্ছারা প্রেম করতে পারে বিয়ে করতে পারেনা
বালের নায়ক... বালের ছবি... বালের পরিচালক... ॥ বালের অভিনয়।	বালের নায়ক বালের ছবি বালের পরিচালক বালের অভিনয়

Table 5 Demographic information of annotators

	AN-1	AN-2	AN-3	AN-4	Expert
Role	Undergraduate	Undergraduate	Research Assistant	Research Assistant	Professor
Research area	NLP	NLP	NLP	NLP	NLP, HCI
Experience	1 year	1 year	2 years	2 years	15 years
Age	22	23	25	24	51
Gender	Male	Female	Male	Female	Male
Religion	Hindu	Islam	Hindu	Islam	Islam

4.3.1 Annotators

HS identification varies from person to person depending on their age, religion, social status [49]. Furthermore, if the annotators are a homogeneous group then it may fail to capture the full range of online abusive contents [50]. As a result, the dataset was manually annotated by five individuals: two undergraduates, two graduates, and one academic expert. Details regarding their experience, expertise, and other relevant demographic information are provided in Table 5. Key characteristics of the undergraduate and graduate annotators include: (a) ages ranging from 22 to 26 years, (b) research focus in NLP with 1–3 years of experience, (c) moderate religious perspectives, (d) no affiliation with political organizations, and (e) active engagement with social media, including awareness of online abusive contents.

4.3.2 Annotation rules

As previously mentioned, we enlisted annotators from various segments of society. Since HS is a subjective matter, our annotators adhered to specific rules during the annotation process to ensure consistency. Our annotators followed the process defined in Algorithm 1 inspired by [25]. The annotation process was divided into two parts. HS recognition and category identification.

```

1: Input: Set of comments with initial labels
2: Output: Final labeled HS dataset
3: Let  $C = \{C_1, C_2, \dots, C_n\}$  be the set of comments (texts)
4: Initialize  $BCD$  (Bangla hate comment dataset) as an empty list
5: Initialize  $FTL$  (Final Target Labels) as an empty list
6: Let  $IL[n][2] = \{a_1, a_2, \dots, a_n\}$  represent the initial labels
7: Initialize  $D$  (disagreement list) as an empty list
8: for each comment  $C_i$  in  $C$  do
9:   Assign  $l1 \leftarrow IL[i][1]$  and  $l2 \leftarrow IL[i][2]$ 
10:  if both  $l1$  and  $l2$  are flagged then
11:    Discard the comment
12:  else if  $l1 = l2$  then
13:    Append  $C_i$  to  $BCD$  and  $l1$  to  $FTL$ 
14:  else
15:    Append  $C_i$  to  $D$  (disagreement)
16:  end if
17: end for
18: for each comment  $d_j$  in  $D$  do
19:   Resolve disagreement with annotators
20:   Either add  $d_j$  to  $BCD$  with its final label or discard it
21: end for

```

Algorithm 1 HS dataset labeling algorithm

1. HS Identification: We categorized any comments as HS if they directly target individuals or specific societal groups. From the analysis of the texts presented in the Table 6, it is evident that comments are classified as HS when they contain direct attacks on individuals or particular groups within society. Conversely, comments that do not contain such attacks are categorized as NHS.
2. Category Identification: Once the annotators identified comments as HS, they further categorized them into five distinct types presented in Table 7. Comments that aligned with these predefined types were labeled accordingly, while those that did not match any of the predefined types were discarded. To enhance clarity and consistency, we have provided our annotators with detailed category definitions, comprehensive examples, and refined our guidelines based on the feedback from preliminary results.

Table 6 HS identification

Comment	HS
রোহিঙ্গারা অনেক অপরাধপ্রবণ এবং তাদের বিশ্বাস করা যায় না Translation: Rohingyas are very prone to crime and cannot be trusted	Yes
আন্তর্জাতিক সহযোগিতা ও বৈচিত্র্য আমাদের অর্থনীতিকে শক্তিশালী করে Translation: International cooperation and diversity strengthen our economy	No
আওয়ামীলীগের সদস্যরা দেশের শত্রু এবং তাদের ধ্বংস করা উচিত Translation: The members of the Awami League are enemies of the country and should be destroyed	Yes

Table 7 Category identification

Comment	Category
রাজনৈতিক নেতারা শুধুমাত্র নিজস্ব স্বার্থের জন্য কাজ করে দেশের উন্নতি তাদের মাথায় নেই Translation: Political leaders only work for their own interests with no regard for the countrys development	PoHS
এই ধর্মীয় সম্প্রদায়ের সদস্যরা কুসংস্কারের বন্দী তাদের সকল ধর্মীয় আচরণ কেবল অন্ধবিশ্বাসের পরিচয় Translation: The members of this religious community are captive to outdated customs all their religious practices are merely a sign of blind superstition	ReHS
আড়ং গোপনে আর্থিক প্রতারণা ও কর্মচারীদের শোষণে লিপ্ত Translation: Arong is secretly engaged in financial fraud and exploitation of its workers	SlaHS
বিদেশীরা এদেশে অপরাধ ছাড়া আর কি করছে Translation: What else are foreigners doing here except committing crimes	XenHS
নারীরা গাড়ি চালাতে পারে না তাদের রাস্তায় নামাই উচিত নয় Translation: Women cant drive they shouldnt even be on the road	MisoHS

Table 8 Kappa scores for binary and multi-class annotations

Category	Class name	Score	Mean
Binary	HS	0.84	0.83
	NHS	0.82	
Multi-Class	MisoHS	0.72	0.69
	XenHS	0.66	
	PoHS	0.71	
	ReHS	0.78	
	SlaHS	0.58	

5 HateCorpBN-XL (Bangla Hate Corpus Extended)

5.1 Annotation quality

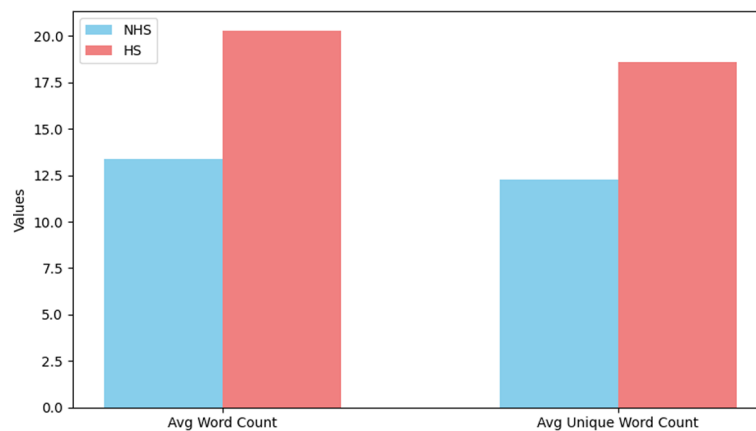
We measured the inter-rater reliability for ensuring the quality of our annotation. We have used kappa score [51] as a measure of reliability of agreement between two annotators who are classifying the comments into different classes. Kappa Score assesses the degree of agreement between two raters (or models) about classification choices while accounting for the likelihood that agreement could arise by chance.

$$k = \frac{Ob(A) - Hyp(A)}{1 - Hyp(A)} \quad (1)$$

Here, Ob(A) Denotes Observed Agreement and Hyp(A) denoted hypothetical agreement among the annotators. Table 8 shows the kappa score for both binary and

Table 9 Dataset category statistics

Category	Total HS
PoHS	6050
ReHS	6050
MisoHS	6050
SlaHS	6050
XenHS	6050

**Fig. 3** Average word count comparison for Task-1

multi-class annotation levels. For binary classes it was observed that the average score is 0.83 which means a perfect agreement between the annotators. In the case of multi-class annotation the highest score of 0.78 was achieved for ReHS, which indicates a substantial agreement. The minimum score of 0.58 was obtained for SlaHS, it indicates a moderate agreement. The average score in multi-class annotation was found 0.69 indicating a substantial agreement. The difference between binary and multi-class annotation scores indicate that the some comments had been communicated subtly so it was hard to classify as a specific class.

5.2 Dataset statistics

The dataset was split into three sets—training (80%), validation (10%), and test (10%)—to make training and evaluation easier. To ensure unpredictability and remove bias, the dataset's instances were randomly shuffled before partitioning. The dataset consists of 65,251 texts in total. Of these, 35,001 messages are classified as NHS, indicating that the content is not aggressive. The remaining 30,250 texts are classified as HS, indicating that the content is hostile. Five distinct categories are used to further categorize the violent texts: misogynist, xenophobic, anti-political, anti-religious and slanderous. Table 9 displays the quantity of comments in these classes. Figure 3 and 4 shows the average statistics of word counts for both Task-1 and Task-2. We calculated the Jaccard similarity between 2,500 most frequent words of each class. The values for binary class and multi-class are represented in Table 10 and 11. For classification the maximum similarity was between the classes PoHS and ReHS and minimum similarity was between XeHS and SlaHS classes.

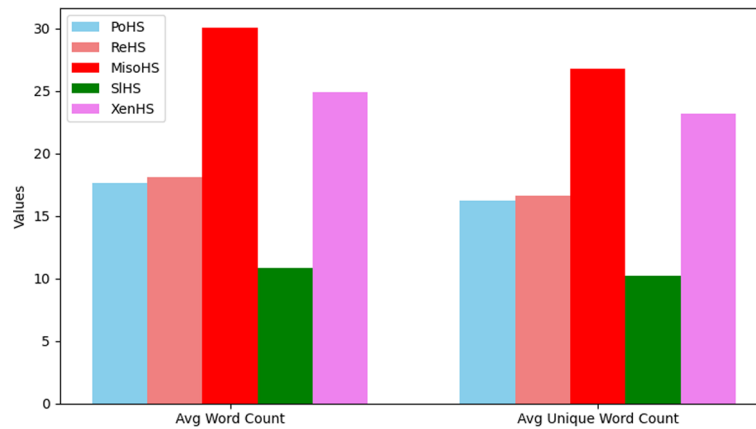


Fig. 4 Average word count comparison for Task-2

Table 10 Jaccard similarity index for binary comments

	NHS	HS
NHS	–	0.2146
HS	–	–

Table 11 Jaccard similarity index for multi-class comments

	PoHS	ReHS	MisoHS	XenHS	SlaHS
PoHS	–	0.41	0.31	0.25	0.42
ReHS	–	–	0.32	0.22	0.43
MisoHS	–	–	–	0.33	0.30
XenHS	–	–	–	–	0.18
SlaHS	–	–	–	–	–

Table 12 Lexical diversity statistics for binary-labeled comments

Label	Total tokens	Unique tokens	TTR
NHS	471,919	47,553	0.1008
HS	616,357	45,896	0.0745

Table 13 Lexical diversity statistics for multi-class hate speech categories

Category	Total tokens	Unique tokens	TTR
PoHS	107,008	15,990	0.1494
ReHS	110,249	12,073	0.1095
MisoHS	181,968	18,063	0.0993
XenHS	151,366	13,601	0.0899
SlaHS	65,813	12,739	0.1936

5.3 Lexical diversity analysis

Tables 12 and 13 present the lexical diversity statistics for the HateCorpBN-XL Bangla hate speech dataset, computed in terms of total tokens, unique tokens, and the Type–Token Ratio (TTR). In the binary setting, the HS class contains 616,357 tokens with a TTR of 0.0745, while the Non-HS class includes 471,919 tokens with a higher TTR of 0.1008. The lower TTR in the HS category indicates a more repetitive and domain-specific vocabulary, often dominated by recurring offensive expressions and group-targeted

terms. In contrast, the Non-HS class exhibits greater lexical variety, reflecting more diverse linguistic contexts.

For the multi-class setting, the misogynist category contains the highest number of tokens (181,968), while slander exhibits the highest TTR (0.1936), suggesting relatively diverse word usage despite a smaller text volume. Conversely, xenophobic (0.0899) and misogynist (0.0993) categories show lower lexical diversity, implying more repetitive language patterns.

6 Methodology

In this section, we present three distinct architectures of HateBertBN model: HateBertBN-CNN, HateBertBN-LSTM, and HateBertBN-MLP and description of the base line transformer model's architecture. Our proposed architectures utilize pre-trained BanglaBERT embeddings and integrate different downstream classifiers to process and classify Bangla text sequences. Each model is designed to effectively capture contextual information from Bangla text while leveraging the strengths of specific neural network layers—CNNs, LSTMs, and MLPs respectively.

6.1 BanglaBERT and other transformer models

The transformer model showed state-of-the-art performance in detecting HS. Several studies [3] have been conducted and proved their efficiency in the performances. Thus, this work employed four pre-trained models: m-BERT [52], Distil-m-BERT [53], BanglaBERT [54] and XLM-R [55]. These models were fine-tuned using our dataset HateCor-pBN-XL, and the model's hyperparameters varied while they had been fetched from huggingface library [56].

As shown in Fig. 5, BanglaBERT follows the architecture of ELECTRA and incorporates 12 layers with 12 attention heads, each with a hidden size of 768, totaling around 110 million parameters. Firstly, bangla text are tokenized and then it feeds into ELECTRA model which consists of two main parts: Electra embeddings and 12 Electra layers. To extract the semantic meaning of the tokens, Electra layers transforms the tokens into

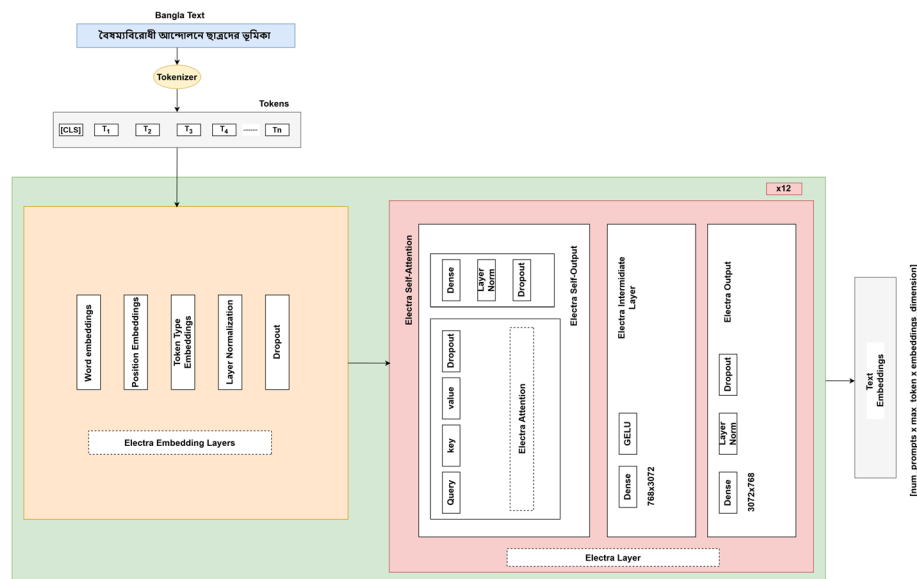


Fig. 5 Architecture of BanglaBERT

continuous vector representations. These representations then passed into Electra layers which consists of three main component Electra Self-Attention, Electra Intermediate, and Electra Output layers. This layers help the model to capture robust contextual information from the given text. By using self attention mechanism via assigning weights to the tokens, it captures internal dependencies. The Electra intermediate layers uses dense layer and GELU (Gaussian Error Linear Unit) activation to represent the contextual information of the given text. Finally, the output of the Electra layer is a text embedding of dimensions of $\text{num_prompts} \times \text{max_length} \times \text{embedding_dim}$. Pre-training was conducted on a large and diverse 18.6 GB corpus containing approximately 8.6 billion tokens, sourced from Bangla Common Crawl, Wikipedia, and news articles. Unlike monolingual models, m-BERT is trained on 104 languages, including Bangla, making it effective for multilingual tasks. In our study, we have used 'bert-base-multilingual-cased' with 12 layers, 12 heads, and 110 M parameters. Distil-M-Bert is a lighter version of BERT, designed to be more efficient while retaining 97% of BERT's performance. We have deployed 'distilbert-base-multilingual-cased' which had 6 transformer layers, half of the original BERT's 12 layers, with the same hidden size of 768 and 12 attention heads, resulting in approximately 66 million parameters. DistilBERT is trained using knowledge distillation on the same corpus as BERT, where it learns from BERT's output, making it faster and more memory-efficient. Another transformer model, 'XLM-R-Base' is also implemented in our work. It is a multilingual transformer model based on the RoBERTa architecture and pre-trained on 2.5TB of filtered CommonCrawl data, covering 100 languages.

6.2 HateBertBN-CNN model

The first architecture, HateBertBN-CNN presented in Fig. 6(a), is designed to extract n-gram features from the text sequence using convolutional layers to capture both global and local features which helps in identifying short, contextually important phrases within a sentence. We do not employ explicit morphological embeddings; rather, we simply leverage BanglaBERT's subword-level tokenization. This subword segmentation enables the model to effectively capture the agglutinative and morphologically rich characteristics of Bengali, providing robustness to vocabulary variation, spelling inconsistencies. It will also reduce the computational cost due to fewer sequential dependencies. Therefore, it will enhance the generalization and robustness of the models compared to other transformers.

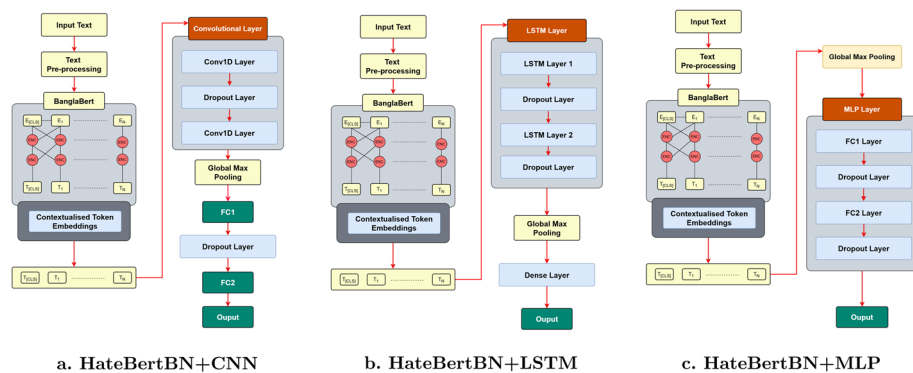


Fig. 6 Architectures of proposed models: **a** HateBertBN+CNN, **b** HateBertBN+LSTM, **c** HateBertBN+MLP

6.2.1 Preparing contextualized embeddings

The process begins with preprocessing the input text sequence $X = [x_1, x_2, \dots, x_T]$. The sequence is tokenized and augmented with the special tokens $[\text{CLS}]$ and $[\text{SEP}]$, yielding:

$$X_{\text{prep}} = [[\text{CLS}], x_1, x_2, \dots, x_T, [\text{SEP}]] \quad (2)$$

Then, these tokens are passed through Electra embedding layers of BanglaBERT. Each token x_i in the preprocessed sequence is mapped to a dense vector through a combination of token, positional, and segment embeddings:

$$E(x_i) = E_{\text{token}}(x_i) + E_{\text{pos}}(i) + E_{\text{type}}(x_i) \quad (3)$$

where $E_{\text{token}}(x_i)$ is the token embedding, $E_{\text{pos}}(i)$ is the positional embedding computed as:

$$E_{\text{pos}}(i) = \begin{cases} \sin\left(\frac{i}{10000^{2k/d_{\text{model}}}}\right), & k \text{ is even,} \\ \cos\left(\frac{i}{10000^{2k/d_{\text{model}}}}\right), & k \text{ is odd} \end{cases} \quad (4)$$

and $E_{\text{type}}(x_i)$ is the segment embedding. Combining these components produces the initial sequence representation:

$$H_0 = [E([\text{CLS}]), E(x_1), \dots, E(x_T), E([\text{SEP}])] \quad (5)$$

This representation is fed into Electra layer of BanglaBERT, a transformer-based encoder with L layers, where each layer applies multi-head self-attention and position-wise feed-forward transformations. The multi-head self-attention mechanism can be represented as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \quad (6)$$

where $Q = H^{(l-1)}W_Q$, $K = H^{(l-1)}W_K$, and $V = H^{(l-1)}W_V$. This is followed by a position-wise feedforward network:

$$\text{FFN}(H') = \text{GELU}(H'W_1 + b_1)W_2 + b_2 \quad (7)$$

with layer normalization and residual connections. Layer normalization helps to stabilize the output which is defined as:

$$\text{LayerNorm}(x) = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \cdot \gamma + \beta, \quad (8)$$

where μ and σ^2 are the mean and variance of x , and γ, β are learned parameters. Followed by the position-wise feedforward network, we combine layer normalization and residual connection:

$$H^{(l)} = \text{LayerNorm}(\text{Attention}(Q, K, V) + H^{(l-1)}) \quad (9)$$

$$H^{(l)} = \text{LayerNorm}(\text{FFN}(H^{(l)}) + H^{(l)}) \quad (10)$$

After $L(12)$ layers which is twelve in our case, the final contextualized embeddings are:

$$H = H^{(L)} \quad (11)$$

6.2.2 CNN-based classification layers

The contextualized embeddings H from Eq. 11 are then processed through a series of one-dimensional convolutional layers to extract n -gram features. The first convolutional layer applies:

$$C_1 = \text{ReLU}(W_1 * H + b_1), \quad (12)$$

and then passed through a dropout layer. The second convolutional layer refines the extracted features:

$$C_2 = \text{ReLU}(W_2 * C_1 + b_2). \quad (13)$$

To generate a fixed-size representation, global max pooling is applied to the output of the final convolutional layer:

$$p = \text{GlobalMaxPooling}(C_2). \quad (14)$$

The pooled vector p is passed through a fully connected layer with ReLU activation:

$$z = \text{ReLU}(W_{fc1}p + b_{fc1}), \quad (15)$$

The output then goes through a dropout layer and another fully connected layer. After that, the final classification logits are obtained using a softmax layer:

$$\hat{y} = \text{Softmax}(W_{fc2}z + b_{fc2}). \quad (16)$$

6.3 HateBertBN-LSTM model

In the second architecture, HateBertBN-LSTM, we use a bidirectional LSTM network to capture long-range dependencies in the text sequence. The key advantage of using LSTM layer is that it preserves the long-range dependencies of a sentence, which transformers alone might struggle with due to positional encoding limitations. It will also help to retrieve the sequential context awareness of a sentences which is ideal for structured narratives and contextual learning. Figure 6(b), illustrates the architecture diagram of HateBertBN-LSTM. Similar to the CNN model, the input sequence is passed through BanglaBERT to produce contextualized embeddings, as described in Eq. 11.

The embeddings H are then fed into a bi-directional LSTM network, generating forward and backward hidden states for each token in the sequence:

$$h_i^{\text{LSTM}} = [\vec{h}_i^{\text{LSTM}}, \overleftarrow{h}_i^{\text{LSTM}}] \quad (17)$$

where $\vec{h}_i^{\text{LSTM}}$ and $\overleftarrow{h}_i^{\text{LSTM}}$ represent the forward and backward hidden states for token i . The combined hidden state has a dimension of $2H$, where H is the LSTM hidden size.

Then the output goes through a dropout layer and another LSTM layer, followed by a dropout layer. Following the LSTM layers, global max pooling is applied to obtain a fixed-size vector representation:

$$p = \text{GlobalMaxPooling}(h_1^{\text{LSTM}}, \dots, h_T^{\text{LSTM}}) \quad (18)$$

The pooled vector p is then passed through dense layer via ReLU activation to generate the final output:

$$z = \text{ReLU}(W_{fc1}p + b_{fc1}) \quad (19)$$

$$\hat{y} = \text{Softmax}(W_{fc2}z + b_{fc2}) \quad (20)$$

6.4 HateBertBN-MLP model

In the final architecture, HateBertBN-MLP, we adopt a multi-layer perceptron (MLP) as the downstream classifier. One of the main advantages of using MLP is that it helps in feature transformation and decision-making from the embeddings of BanglaBERT. The global features extracted by BanglaBERT are effectively mapped to classification labels via MLP as a downstream classifier. It is also lightweight and fast compared to other transformer models. As shown in Fig. 6(c), the input sequence is passed through BanglaBERT, yielding contextualized token embeddings H , as described earlier in Eq. 11.

Instead of using convolutional or LSTM layers, global max pooling is applied directly to the embeddings:

$$p = \text{GlobalMaxPooling}(H) \quad (21)$$

The pooled vector p is then passed through a series of fully connected layers with ReLU activation, utilizing dropout layer in between them:

$$z_1 = \text{ReLU}(W_{fc1}p + b_{fc1}) \quad (22)$$

$$z_2 = \text{ReLU}(W_{fc2}z_1 + b_{fc2}) \quad (23)$$

Finally, the output layer applies softmax to produce the predicted class probabilities:

$$\hat{y} = \text{Softmax}(W_{out}z_2 + b_{out}) \quad (24)$$

6.5 Hyperparameters

All of our pretrained transformer models has been fine tuned using our proposed HateCorpBN-XL dataset. We have presented the hyperparameters for our models in Table 14. 15 epochs with a batch size of 8 has been used for the training purpose of all our models. During training, the model is optimized using the Adam optimizer, which adjusts the learning rate dynamically. Additionally, a learning rate scheduler is employed with a maximum learning rate of $2e^{-5}$ to reduce the learning rate when the validation accuracy plateaus, thereby enhancing the model's convergence. The cross-entropy loss function is used to measure the discrepancy between the predicted logits and the actual

Table 14 Hyperparameters list

Hyperparameters	Value
Number of classes	2, 5
Batch size	8
Learning rate	2×10^{-5}
Optimizer	Adam
Loss function	CrossEntropyLoss
Dropout rate	0.3
Epochs	15
Max sequence length	50, 60

labels, guiding the backpropagation process to update the model's parameters. The overall performance is monitored through training and validation accuracy metrics. We have used max sequence length for texts is adjusted to 50 for Task-1, 60 for Task-2. We also have kept similar hyparameters and policy for our proposed models as well.

7 Result analysis

7.1 Experimental configuration and evaluation metrics

Python 3 was used in experiments with a Google Cloud Engine backend (GPU) on the Google Colaboratory platform. 78 GB of disk space and 32 GB of RAM were used to implement the models. We processed and prepared the data using numpy (2.3.0) and pandas (2.3.0). While the transformer models were implemented trained using PyTorch (2.7.1).

The models are developed using the train, validation, and test instances, guaranteeing that every instance in these sets is exclusive of every other occurrence. While the validation set is used to determine the hyperparameter values, the models learn from the cases in the training set. Lastly, the test set's unseen cases are used to assess the trained models. The systems' performances are computed and compared using a variety of statistical metrics. The evaluation metrics that were employed are depicted in Eqs. 25 to 30.

1. Precision: Precision measures the proportion of correctly classified instances or samples among those labeled as positives.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (25)$$

2. Recall: Recall is a metric that measures the frequency at which a machine learning model correctly identifies positive instances (true positives) from the total actual positive samples in the dataset.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negative}} \quad (26)$$

3. Error: This measures of the frequency of wrongly classified values.

$$\text{Error} = \frac{\text{False Positives} + \text{False Negative}}{\text{Number of Samples}} \quad (27)$$

4. F_1 -Score: This the simple average of precision and recall.

$$F = \frac{2PR}{P + R} \quad (28)$$

5. Weighted F_1 -Score: Weighted F_1 Score is calculated on the imbalanced dataset.

$$WF = \frac{1}{N} \sum_{i=1}^k F_i n_i, N = \sum_{i=1}^k n_i \quad (29)$$

Here N , F_i and n_i indicates total samples in test set, f_1 -score and number of samples in class(i).

6. Macro F_1 -Score: The macro-averaged F1 score (or macro F1 score) is calculated by taking the arithmetic mean (unweighted mean) of the F1 scores for each class.

$$\text{MacroF} = \frac{\sum_{i=1}^n F_i}{n} \quad (30)$$

Here F_i indicates F_1 score of class(i) and n indicates number of classes.

7.2 Results

7.2.1 Task-1 performance analysis

Table 15 provides a comparative analysis of the performance of several classifiers in detecting NHS and HS. The BanglaBERT and XLM-R-Base models demonstrated similar performance, attaining a total accuracy of 0.91. XLM-R-Base performed slightly better on HS classification (precision of 0.89) than BanglaBert. M-BERT demonstrated competitive results with an accuracy of 0.90, performing well across both categories. Distill-M-BERT, although slightly behind with an accuracy of 0.87. Its precision and F1-scores for NHS and HS were slightly lower compared to the other models. The proposed three variants of the HateBertBN model outperformed all state-of-the-art models by attaining the highest overall accuracy of 0.92. Among our three variants, HateBertBN-CNN and HateBertBN-MLP showed superior performance in both NHS and HS detection tasks. In terms of recall value, HateBertBN-MLP (recall of 0.92 for HS) is the highest-performing model, while HateBertBN-CNN achieved the lowest recall value of 0.91. HateBertBN-LSTM also showed promising results for NHS by achieving a 0.93 precision value, but it suffered in the HS detection task as it achieved only 0.89 precision, which is the lowest compared to the other three variants. We have observed similar consistent and high F1-score 0.92, 0.91 for NHS, HS respectively among three variants of HateBertBN models. Moreover, from the confusion matrix Fig. 7 (a), (b), (c) and jaccard similarity Table 10, our best performing models results can be justified since there are less misclassification rate with low jaccard similarity score between the classes.

7.2.2 Task-2 performance analysis

For Task two, we also evaluated the proposed transformer models and compared our results with the current state-of-the-art. Table 16 depicts the detailed results of each

Table 15 Evaluation results of Task-1 on test set

Classifier	Class	Precision	Recall	F_1 -score	Accuracy
BanglaBERT	NHS	0.93	0.89	0.91	0.91
	HS	0.88	0.93	0.90	
Distill-M-BERT	NHS	0.90	0.85	0.88	0.87
	HS	0.84	0.89	0.87	
M-BERT	NHS	0.91	0.90	0.90	0.90
	HS	0.88	0.90	0.89	
XLM-R-Base	NHS	0.93	0.90	0.92	0.91
	HS	0.89	0.92	0.91	
HateBertBN-MLP	NHS	0.93	0.92	0.92	0.92
	HS	0.91	0.92	0.91	
HateBertBN-LSTM	NHS	0.93	0.90	0.92	0.92
	HS	0.89	0.93	0.91	
HateBertBN-CNN	NHS	0.92	0.92	0.92	0.92
	HS	0.91	0.91	0.91	

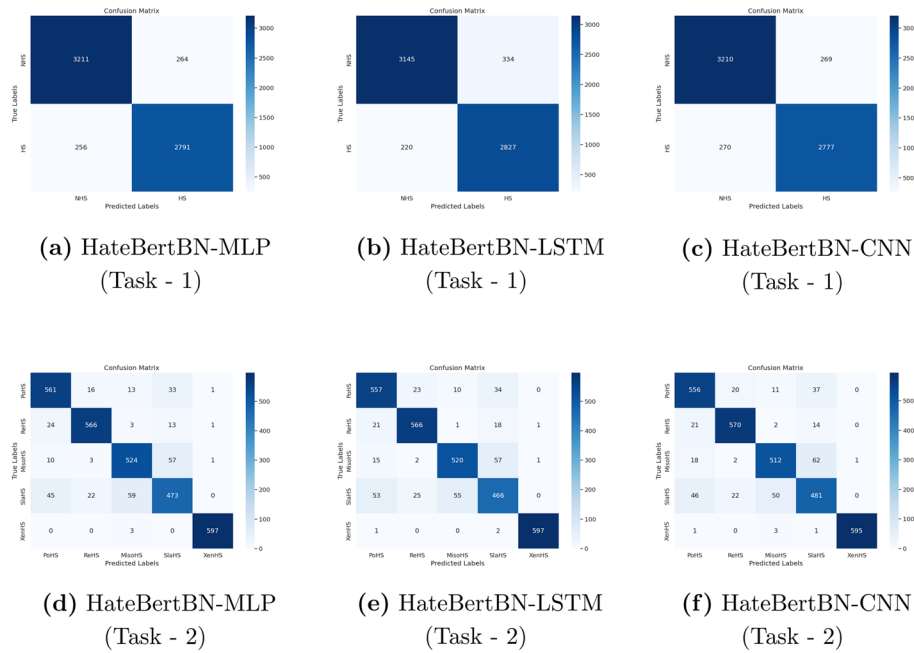


Fig. 7 Confusion matrices for HateBertBN models. **a** HateBertBN-MLP (Task-1), **b** HateBertBN-LSTM (Task-1), **c** HateBertBN-CNN (Task-1), **d** HateBertBN-MLP (Task-2), **e** HateBertBN-LSTM (Task-2), and **f** HateBertBN-CNN (Task-2)

model. The two variants of HateBertBN (HateBertBN-CNN, HateBertBN-MLP) models consistently outperformed the others, achieving the highest accuracy of 0.90, while HateBertBN-LSTM achieved low accuracy of 0.89 among the three variants. Distill-M-BERT has the weakest performance, with an accuracy of 0.87 and poor precision and F1-scores, especially in PoHS. M-BERT struggles in SlaHS classification with a recall of just 0.68. XLM-R-Base performs better than Distill-M-BERT; it falls short in MisoHS with a precision of 0.83. BanglaBERT performed similarly to XLM-R-Base with an accuracy of 0.89 but showed weaknesses in SlaHS. M-Bert achieved a low accuracy and F1-score of 0.87. Despite its poor performance, M-Bert showed a slightly improved precision of 0.97 in XenHS compared to Distill-M-Bert. In the case of MisoHS, Distill-M-BERT achieved the highest precision of 0.85 compared to other state-of-the-art models. In contrast, HateBertBN-MLP and HateBertBN-CNN demonstrated balanced precision and recall across all categories for Task-2. While HateBertBN-MLP excels in ReHS with the highest precision of 0.93, HateBertBN-CNN achieved the best precision of 1.00 and F1-score 1.00 in XenHS. Besides that, HateBertBN-MLP gained the highest precision score of 0.82 in SlaHS among the three variants. Compared to the other two variants, HateBertBN-LSTM achieved the lowest accuracy of 0.89. It also has the lowest precision of 0.81 in SlaHS among them. In the case of MisoHS, HateBertBN-LSTM showed the highest precision of 0.89 among all the proposed models. In terms of the F1-score, HateBertBN-MLP showed the most balanced performance across all the classes. In general, our models tend to misclassify SlaHS, MisoHS and PoHS class. The confusion matrix Fig. 7d–f for Task-2 and jaccard similarity Table 11 for multiclass label in the dataset reveals that there are similarities among the classes SlaHS, MisoHS and PoHS. As a result, our models mostly suffered in those classes. Overall, HateBert-CNN and HateBert-MLP are the best performing model for Task-2.

Table 16 Evaluation results of Task-2 on test set

Classifier	Class	Precision	Recall	F_1 -Score	Accuracy
BanglaBERT	PoHS	0.89	0.86	0.88	0.89
	ReHS	0.91	0.96	0.93	
	MisoHS	0.84	0.89	0.86	
	SlaHS	0.83	0.77	0.80	
	XenHS	1.00	0.98	0.99	
Distill-M-BERT	PoHS	0.82	0.89	0.85	0.87
	ReHS	0.91	0.94	0.92	
	MisoHS	0.85	0.80	0.82	
	SlaHS	0.81	0.75	0.77	
	XenHS	0.96	0.97	0.97	
M-BERT	PoHS	0.87	0.85	0.86	0.87
	ReHS	0.87	0.97	0.92	
	MisoHS	0.80	0.88	0.84	
	SlaHS	0.83	0.68	0.75	
	XenHS	0.97	0.97	0.97	
XLM-R-Base	PoHS	0.88	0.90	0.89	0.89
	ReHS	0.91	0.93	0.92	
	MisoHS	0.83	0.88	0.86	
	SlaHS	0.83	0.78	0.81	
	XenHS	1.00	0.96	0.98	
HateBertBN-MLP	PoHS	0.88	0.90	0.89	0.90
	ReHS	0.93	0.93	0.93	
	MisoHS	0.87	0.88	0.88	
	SlaHS	0.82	0.79	0.81	
	XenHS	0.99	0.99	0.99	
HateBertBN-LSTM	PoHS	0.86	0.89	0.88	0.89
	ReHS	0.92	0.93	0.93	
	MisoHS	0.89	0.87	0.88	
	SlaHS	0.81	0.78	0.79	
	XenHS	1.00	0.99	1.00	
HateBertBN-CNN	PoHS	0.86	0.89	0.87	0.90
	ReHS	0.92	0.94	0.93	
	MisoHS	0.88	0.88	0.88	
	SlaHS	0.84	0.78	0.81	
	XenHS	1.00	0.99	1.00	

7.3 Error analysis

We have already established that our proposed HateBertBN models outperformed all others in the Bangla HS classification task. To further strengthen our findings, we have presented quantitative and qualitative error analysis for each of our tasks to acquire more detailed insights into the performance of proposed models.

7.3.1 Quantitative analysis

Figure 7 illustrate the classification matrices of proposed HateBertBN architectures for Task-1 and Task-2. For our task one, all three models- Figure 7(a) HateBertBN-MLP, Figure 7(b) HateBertBN-LSTM, and Figure 7(c) HateBertBN-CNN demonstrated comparable abilities to distinguish between NHS and HS detection. However, the HateBertBN-LSTM model stands out with fewer HS misclassifications, registering 220 false negatives compared to HateBertBN-MLP's 256 and HateBertBN-CNN's 270. On the other hand, for NHS detection HateBertBN-MLP's has the lowest false positive of 264 compared to other variants. Sometimes, the sentiment of a sentence changes due to the

implicit use of hate words. As a result, they don't express hatred explicitly, and those sentences are difficult to classify.

In Task-2, HateBertBN-MLP Figure 7(d), scored fewer misclassifications across HS categories like PoHS, MisoHS compared to other two models. It classified 561 and 524 instances correctly among 624 and 595 instances of PoHS, MisoHS classes. Then HateBertBN-CNN, Fig. 7(f) showed lowest misclassification tendency for ReHS and SlaHS classes which is also highest among the three models. It was able to detect 570 and 481 instances positively among 607 and 599 instances of ReHS, SlaHS classes. Although, HateBertBN-LSTM classified 566 and 597 instances correctly among 607 and 600 instances of ReHS, XeHS classes, it performed poorly in most categories demonstrated in Fig. 7(e). Interestingly, all three model performed well in detecting XeHS class but suffered mostly in detecting SlaHS class. This is because the presence of similar slanderous words may present in all HS classes which changes the HS categories based on different context.

The error rates reported in Table 17 further corroborate the model performance across two tasks. HateBertBN-MLP achieves the lowest error rate in Task-1 at 8.03%, closely followed by HateBertBN-CNN and LSTM, which record 8.26 and 8.49%, respectively. Similarly, in Task-2, HateBertBN-MLP continues to excel with the lowest error rate of 10.05%, while CNN slightly surpasses LSTM, with error rates of 10.08 and 10.55%, respectively. On the other hand, Distil-M-Bert and M-Bert struggle with noticeably higher error rates, particularly in Task-2, where their errors exceed 13%. While XLM-R demonstrates relatively strong performance, its error rates (8.95% in Task-1 and 11.01% in Task-2) still do not match the efficiency of the HateBertBN variants, especially HateBertBN-MLP and HateBertBN-CNN.

In summary, for Task-1, HateBertBN-LSTM demonstrates the lowest false negatives for HS detection, while HateBertBN-MLP shows the least false positives for NHS, indicating its precision in binary classification. In Task-2, HateBertBN-MLP excels in correctly classifying PoHS and MisoHS instances, while HateBertBN-CNN shows superior performance in ReHS and SlaHS categories, minimizing misclassifications. Interestingly, all three models handle the XeHS class effectively but struggle with SlaHS due to contextual nuances. The error rates further confirm HateBertBN-MLP's consistent performance, achieving the lowest rates across both tasks, with HateBertBN-CNN slightly trailing behind. In contrast, models like Distil-M-Bert and M-Bert underperform with significantly higher error rates, solidifying the HateBertBN variants, particularly MLP and CNN, as the most effective for HS detection.

Table 17 Error rate of proposed models

Model	Task 1	Task 2
BanglaBERT	9.29%	10.64%
Distil-M-BERT	12.96%	13.02%
M-BERT	10.10%	13.22%
XLM-R	8.95%	11.01%
HateBertBN-MLP	8.03%	10.05%
HateBertBN-LSTM	8.49%	10.55%
HateBertBN-CNN	8.26%	10.08%

Table 18 Missed classification examples for Task-1

Comment	Actual	MLP	CNN	LSTM
এত খারাপ মানুষ ওকে ডিম থেরাপি দিতে হবে As he is a bad guy, he should be given egg therapy	HS	HS	NHS	NHS
ভারত না পাকিস্তান টাইপে ভুল হইসে There is a spelling mistake between India and Pakistan	NHS	HS	NHS	HS
আমি এই সরকারের কাজকর্মে বিরক্ত I'm frustrated with this government's actions	NHS	HS	HS	NHS

Table 19 Missed classification examples for Task-2

Comment	Actual	MLP	CNN	LSTM
ওই জামাতি গান্ধারগুলো সব পাকিস্তানি এজেন্ট Those Jamaati traitors are all Pakistani agents	PoHS	SlaHS	PoHS	XenHS
রুমিন ফারহানার প্রমোশনটা ঘুষ আর অফিসে বসের সঙ্গে বিশেষ সম্পর্কের ফল Rumin Farhana's promotion was the result of bribes and a special relationship with the boss	SlaHS	SlaHS	MisoHS	MisoHS
ইস্কন দেশকে বেচে ভারতের দালালি করে ISKCON is selling out the country and working as agents for India	ReHS	ReHS	SlaHS	XeHS

7.3.2 Qualitative analysis

We have further investigated our wrong classification both for Task-1 and Task-2 using the proposed model. Tables 18, 19 depicts the misclassification samples for both tasks. To simplify the table presentation, we have considered HateBertBN-MLP as MLP, HateBertBN-CNN as CNN, HateBertBN-LSTM as LSTM. In some cases, our model accurately classified the HS and its category; at the same time, it misclassified a few samples. Compared to our proposed model, other transformer models also performed similarly. No models could accurately classify all the text into its correct labels. We examined the inaccurate classification, it revealed that in some cases it is difficult to pinpoint words that express or subtly encourage of hatred. Since these occurrences don't use any hostile language or references, it's challenging to highlight them. However, the model incorrectly labels certain texts as hatred even when they utilize sarcastic language and have no malicious intent. Without knowing the context, it is difficult to recognize and categorize such text samples from the surface-level analysis. We also explored the possibilities of comments overlapping but no comments overlapping was found in this case. Certain terms are also commonly used in both HS and NHS classes, such words may cause confusion and add complexity to the tasks when they appear in a text. Incorporating context-aware mechanism and external knowledge sources like HS lexicons or could enhance the model's understanding of subtle and sarcastic expressions. Expanding the dataset with more diverse and nuanced examples might improve the proposed model's classification performance.

7.3.3 Limitations

Despite their strong performance and state-of-the-art results, the proposed HateBertBN models (MLP, CNN, and LSTM) have certain limitations. Categories with high lexical diversity or subtle contextual cues, such as slander, remain challenging, and MLP and CNN may struggle to generalize due to shallow feature fusion. While LSTM captures sequential dependencies more effectively, it can still underperform on complex linguistic constructs like sarcasm, idiomatic expressions, or code-mixed English-Bangla content. The models primarily rely on BanglaBERT embeddings, which may not fully capture pragmatic or socio-cultural nuances. Additionally, sensitivity to class imbalance and repetitive vocabulary (e.g., xenophobic) could lead to overfitting or biased predictions. Addressing these limitations in future work offers opportunities to further enhance robustness, fairness, and generalizability.

7.4 Computational complexity analysis

We conducted a comparative analysis based on the key performance metrics including the number of parameters (millions), inference time (ms/sample), floating-point operations (FLOPs in GFlops), memory usage (MB), model size (MB), and throughput using a batch size of eight (samples/s) to evaluate the computational efficiency of our proposed models. These metrics were measured using PyTorch profiling tools and benchmarked on a high-performance GPU setup provided by Google Colab. Table 20 illustrates the detail comparison of average computational complexity analysis for both tasks.

Our proposed HateBert models (CNN, LSTM, MLP) exhibit a lower parameter count compared to m-BERT (177.86M) and XLM-R (278.05M), with HateBert-MLP (110.42M) being the most parameter-efficient. HateBert-MLP also achieves the fastest inference time (12.06 ms/sample), outperforming Distill-m-BERT (17.15 ms/sample) and XLM-R (21.5 ms/sample), while BanglaBERT (686.35 samples/s) and HateBert-LSTM (663.32 samples/s) lead in throughput. Most models maintain similar FLOP counts (34.38 – 35.17 GFlops), except for Distill-m-BERT (17.19 GFlops). Memory consumption varies, with XLM-R consuming the highest (1060.67 MB), while HateBert-MLP (3740.24 MB) and HateBert-CNN (3746.09 MB) achieve efficient utilization. Overall, HateBert-MLP and HateBert-CNN strike an optimal balance between accuracy and computational efficiency, making them suitable for real-time, resource-constrained environments.

Table 20 Computational complexity analysis of different models

Model	Parameters (M)	Inference time (ms/sample)	FLOPs (GFlops)	Memory usage (MB)	Model size (MB)	Throughput (samples/s)
BanglaBERT	110.03	11.66	34.38	3794.95	419.73	686.35
M-BERT	177.86	16.75	34.38	4473.61	678.46	497.63
Distill-M-BERT	134.74	17.15	17.19	4987.85	513.98	320.89
XLM-R	278.05	21.50	34.38	4991.45	1060.67	494.26
HateBertBN-CNN	112.60	12.83	35.17	3746.09	428.23	623.72
HateBertBN-LSTM	121.58	14.17	34.38	3781.51	463.80	663.32
HateBertBN-MLP	110.42	12.06	34.38	3740.24	419.73	565.16

Table 21 Comparison between proposed models and existing work using our dataset

Authors	Methods	F1-score (Task 1/Task 2)
Karim et al. [9]	MConv-LSTM	0.88/0.86
Sarkar et al. [57]	f-BERT	0.90/0.87
Jahan et al. [24]	BanglaHateBERT	0.91/0.88
Sharif et al. [25]	Ensemble of transformer models	0.90/0.88
Keya et al. [30]	G-BERT	0.90/0.88
Sristy et al. [58]	Seq. LSTM	0.90/0.89
Mnassri et al. [59]	Ensemble of BERT	0.89/0.87
Ours	HateBertBN	0.92/0.90

7.5 Comparison between proposed and existing work

We have compared our proposed models and their variants with several existing works. For consistency, previous methods [9, 24, 25, 30, 57–59], are fine-tuned using our dataset and compared their performance with the proposed techniques.

Table 21 shows the comparison of our proposed models performances with other existing methods from the literature. Notably, the MConv-LSTM model from Karim et al. [9] achieved F1-scores of 0.88 and 0.86, while Sarkar et al. [57] proposed f-BERT which performed slightly better with scores of 0.90 and 0.87. Jahan et al. [24] and Keya et al. [30], introduced BanglaHateBERT, G-BERT respectively, achieved strong performances as well, with F1-scores of 0.91, 0.88 and 0.90, 0.88. The ensemble BERT models from Mnassri et al. [59] also delivered competitive results, they still fell short of our proposed model's overall performance. Similarly, we applied the weighted average ensemble transformer models from Sharif [25], but it did not achieve results comparable to ours. Therefore, HateBertBN outperformed the existing methods with F1-scores of 0.92 and 0.90 for Task-1 and Task-2.

8 Conclusion

In this study, we introduced a novel manually annotated Bangla HS dataset, "HateCorpBN-XL", which contains 65,251 text corpora collected from various online social media, and proposed the HateBertBN model, where we have introduced three variants of the model. Our models were evaluated on two tasks where: task one was to identify whether it is HS or NHS, and task two was, for a given HS text, to classify the categories of HS. For task one, there are only two classes—HS, NHS and for task two, there are five classes: PoHS, ReHS, MisoHS, SlaHS and XenHS. To compare the efficiency of our proposed models, we have utilized other transformer (m-BERT, Distil-M-Bert, BanglaBERT, XLM-R) models on our HateCorpBN-XL dataset. In the case of task one, all variants of HateBertBN models have outperformed the current transformer models by achieving the highest accuracy of 0.92 and weighted average F1-score of 0.92. Again, for task two, two of the variants of the proposed model, HateBertBN-MLP, HateBertBN-CNN achieved a superior accuracy of 0.90 and a weighted average F1-score of 0.90. Although our proposed HateBertBN-LSTM has underperformed, it has achieved a remarkable F1-score in Miso-HS (0.88) and XenHS (1.00) classes compared to m-BERT, Distil-m-Bert, HateBertBN, XLM-R-Base models. We have also compared our model performances with existing works, and our proposed models have outperformed them as well. Our error analysis highlights persistent challenges, including implicitly expressed or sarcastic HS and categories with high lexical diversity, such as slander, which are difficult for models to generalize. Repetitive categories, like xenophobic, may also lead

to overfitting. Limitations include reliance on text-only inputs and sensitivity to class imbalance, potentially affecting robustness and fairness. In future, this study plans to diversify the dataset by creating more intricate labels which will further validate the robustness and generalizability of our proposed method. Future work will aim to expand the dataset with more detailed labels and include multimodal information from images and videos. Adding demographic or community context can also make the system fairer and easier to interpret, leading to more reliable HS detection. Exploring transformer models like Longformer or BigBird is another promising direction to improve scalability and handle longer Bengali texts more effectively.

Author contributions

T.A. (Tanvir Azhar) conceptualized the study, supervised the dataset development process, developed and implemented the codebase, proposed model architecture design, conducted all experiments, performed quantitative and qualitative result analysis, generated visualization & figure drawing, table construction, wrote dataset development section, performed hyperparameter tuning, fine tuning the pretrained models, performed & wrote computational analysis sections, wrote the main draft of the manuscript, performed full revision of the manuscript and supervised the entire project. T.M. (Tahsin Mahmud) curated and preprocessed the dataset, coordinated the manual annotation process, and assisted with data acquisition and cleaning. Also contributed in writing the introduction, related works, table & figure editing. M.A.H. (Muhammad Asif Hasan) contributed to dataset development and helped with proof reading the paper via validating the experimental findings. Also contributed in writing the comparison with proposed and existing work. M.N.U. (Mohammed Nazim Uddin) contributed to improve quantitative and qualitative result analysis, and validated experimental findings. Also provided critical supervision, revision to improve the quality of the research work. S.B.P. (Seung-Bo Park) provided senior supervision, project administration, and critical revision of the manuscript for important intellectual content. All authors reviewed and approved the final version of the manuscript.

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Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Code availability

Manuscript code is available at the following GitHub repository: https://github.com/Transhumanistx/HateBertBN_Bangla_HateSpeech_detection

Materials availability

Not applicable.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Human and animal participants statement

None of the authors conducted any experiments with human participants or animals for this work.

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